EBU5204 Operating Systems 2023/24

LAB1: Using Linux Commands

Instructions: Please read the lab sheet carefully attempt what is asked and provide answers on the answer sheet provided to you. This lab will not be assessed but.

Aims

The aim of this lab sheet is to use basic Linux commands to manipulate files and directories. Depending on how familiar you are with these commands you may need to do more, or less, research into what these commands do.

Why do I need to do this?

Because all labs for the module will require you to be able to use the Linux OS. Being familiar with Linux / Unix based Operating Systems is a skill that employers value. You will not need to "remember" all the commands and options for the exam, but many of these (e.g. ls, cd, pwd, cp, etc.) will be essential for "survival" in the rest of the module.

Command Argument and Options

Commands have arguments: command arg1 arg2
Arguments are usually separated by spaces.
A command can also have an option (or 'flag' or 'switch'): command –opt arg1 arg2
The options usually come first. Options are introduced with a '-'.

Getting Help with commands

You can get help in the following ways:

- Follow a tutorial on the internet such as on https://linuxcommand.org/
- Look at https://linux.die.net/
- Try any / all of the following (where command is any Linux command – don't just type command in the shell expecting something to happen!!)

You can also find the **man pages** for commands online, e.g. search for something like "linux man pages

Step 1: Linux Commands for File Manipulation

Open a terminal window in Linux. Look up the following Linux command in the 'man' pages (e.g. typing man cat will display the man pages for the command cat)

cat, cd, pwd, mkdir, rm, rmdir, link, cp, mv, more, head, tail, chmod

Try each command (take care when using rm and rmdir – they can wipe out files / entire folder structures with subfolders). Write a brief description of each command (1-2 sentences max) (on the answer sheet).

Step 2: Using 'ls'

The 'ls' command is complex and has many options. Read the man page (man ls).
Try 'ls –l' which uses the option 'l' (the letter before 'm', not a digit) and gives lines such as:

-rwxr--r--. 1 flakhani staffphdguest 949 Oct 5 10:16 Fork.c

The man pages do not fully explain all the output and options (though there are instructions on getting more detailed documentation at the end of the man page entry). Try experimenting to answer the following questions:

1. The number '949' in the example above is the file size. What are the units?
2. There is a date given – in the example above 'Oct 5, 10:16'. What happened at this date / time?
3. The format of the date can vary (e.g. try to use the command in a directory where you have some files from e.g. Year 1). Explain the two formats and when they are used.

Now use the ls command to list the files inside one of the directories (folders) that you have under your home directory but run the command from your home directory.

Answer the following question:

4. What is the command you issued to list the contents of that directory? What is the argument for the command?

Step 3: Copying Files

Use Linux commands to create a new directory in your file space (e.g. make a new directory EBU5204 in your home directory). Then copy the files from any of your directories to the new directory that you created. Record the commands you used. Using the appropriate 'ls' command options / flags, determine:

- What information about the files is preserved by the copy?
- What information, if any, is changed

Step 4: Using Pipes

A Linux 'pipe' passes data output by one process to the input of another process. It is like a temporary and invisible file between the processes. The symbol '|' is used to represent a pipe.

Examples

Try the following examples:

ls | wc

ls –l | wc (note that l is the letter)

In each case explain what is happening (you may have to read the man page on 'wc').

Sorting the output of ls

The Linux program ‘sort’ can sort a file, or the output from another program. The aim of this task is to use sort to sort the output of the ls command:

1. by file name
2. by file size

Note: there are options of the ‘ls’ command to do this anyway, but using them is not the aim of the exercise. Instead:

- Choose options for the ‘ls’ command to show size
- Choose options for the ‘sort’ command to control sorting
- Combine using a pipe

Step 5 – File access permissions

We learn about this topic during the Block 2 teaching (File Systems). Files and directories in Linux have access permissions for the owner, owner’s user group and the rest of the users. In the example used to describe file access permissions in Linux example of the output of ls in Step 2 we saw:

-rwxr--r--. 1 flakhani staffphdguest 949 Oct 5 10:16 Fork.c

- Recall your knowledge and describe what the access permissions are for this file for the owner, his/her user group and for the rest of the users.
- The chmod command can be used to change the access permissions for files / directories.
- In the example output before, what effect would the following command have?

  chmod 644 Fork.c

  Briefly explain the numerical coding